



Cambridge (CIE) IGCSE Computer Science



Your notes

Data Storage

Contents

- * Primary Storage
- * Secondary Storage
- * Virtual Memory
- * Cloud Storage



Your notes

Primary Storage

Primary Storage

- Primary storage is directly accessed by the CPU
- Computer systems need both primary and secondary storage to operate
- Both types of storage play a crucial role in the operation of a computer system
- A quick comparison of primary and secondary storage shows:

Primary	Secondary
Volatile (with the exception of ROM)	Non-volatile
Small capacity	Large capacity

Why do you need primary storage?

- A computer needs primary storage because **access times** are considerably **faster than secondary**
- This means the **time taken** to complete operations such as the **Fetch-Execute Cycle** is **dramatically reduced**
- Primary storage holds the **data** and **instructions** that the CPU needs to access whilst the computer is turned on
- Due to the fast access times, primary storage is used as **short-term, working memory**, in hardware that is **directly connected to the CPU** such as **RAM**, and components that reside inside the CPU such as **Cache** and **Registers**
- Performance of primary storage means a much higher **cost** which limits the amount that is used
- For example, **RAM** is commonly purchased in 16 or 32 **gigabytes** whereas secondary storage such as a **hard drive** is in **terabytes**

RAM

What is RAM?

- RAM (Random Access Memory) is **primary storage** that is directly connected to the CPU and holds the **data** and **instructions** that are **currently in use**
- RAM is **volatile** which means the contents of RAM are lost when the power is turned off



Your notes

- For the CPU to access the data and instructions they must be **copied from secondary storage**
- RAM is **very fast** working memory, much faster than secondary storage
- RAM is **read/write** which means data can be read from and written to
- In comparison to ROM, it has a much **larger capacity**



Examiner Tips and Tricks

Just knowing what RAM is and what it stores is not enough, you must be able to apply your understanding of RAM to real world contexts where RAM is a component, e.g. TVs, game consoles and other smart devices



Worked Example

A smart television allows the user to search the Internet and watch videos online.

The smart television uses RAM

Give **two examples** of data that the smart television could store in RAM [2]

How to answer this question

- Think about the main function of a smart television, watch channels, use apps to stream content and browse the web etc
- For each function, try to think of what data would have to be in the RAM whilst you were actually doing it (**in use!**)

Possible answers

- Current channel being watched
- Current volume
- Current video/file/tv program being watched
- Web browser/applications that are **running**
- Data being downloaded/buffered



Examiner Tips and Tricks



Your notes

Examiners will accept anything reasonable when answering these style of questions **BUT** brand names will not be awarded without giving an example

ROM

What is ROM?

- ROM (Read Only Memory) is **primary storage** that holds the **first instructions** a computer needs to **start up** (Bootstrap)
- ROM contains the **BIOS** (Basic Input Output System)
- ROM is a small **memory chip** located on the computers **motherboard**
- ROM is **fast** memory, much faster than secondary storage but **slower than RAM**
- ROM is **non-volatile** which means the contents of ROM are not lost when the power is turned off
- ROM is **read only** which means data can only be read from
- In comparison to RAM, it has a much **smaller capacity**



Examiner Tips and Tricks

RAM & ROM are examples of primary storage, they can be referred to as Main Memory or Primary Memory in the exam



Worked Example

Quinn has 512 kilobytes of ROM and 16 gigabytes of RAM in her computer

1. Describe the purpose of the ROM in Quinn's computer [2]
2. Describe the purpose of the RAM in Quinn's computer [2]

Answer

1. ROM
 - Store start-up instructions/bootstrap
 - Used to start the computer
2. RAM
 - Stores the parts of the OS / programs that are running

- Stores data currently in use
- ...for access by the CPU

Guidance

- Do not confuse the purpose with characteristics, describe **what it does**, not **what it is**



Your notes



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Secondary Storage

Secondary Storage

- Storage devices are **non-volatile secondary storage**, that retain digital data within a computer system
- They provide a means of storing, accessing, and retrieving data, which can include software applications, documents, images, videos, and more
- There are 3 types of storage:
 - Magnetic**
 - Solid-state (flash memory)**
 - Optical**
- Computer systems need both primary and secondary storage to operate
- Both types of storage play a crucial role in the operation of a computer system
- A quick comparison of primary and secondary storage shows:

Primary	Secondary
Volatile (with the exception of ROM)	Non-volatile
Small capacity	Large capacity

Why do you need secondary storage?

- A computer needs secondary storage for **long term storage** of programs and data that are **currently not in use**
- Secondary storage is needed as ROM is read only and RAM is volatile
- Secondary storage holds the **programs** and **data** whilst the computer is turned off (**non-volatile**)
- Performance of secondary storage is **slower** than primary storage but **capacity** is much **higher** which makes it perfect for backup & archive of data files

What are the characteristics of secondary storage?

- Capacity** - What is the maximum amount of data that can be stored?
- Speed** - How fast can data be read from and written to? (R/W)



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- **Cost** - How much does it cost?
- **Portability** - How easy is it to move around? What is the physical size? Weight?
- **Durability** - How robust is the storage?
- **Reliability** - What is the probability that the device will consistently perform its function?



Worked Example

A games console has secondary storage.

State, using an example, why the games console needs secondary storage [2]

How to answer this question

- The keyword is '**why**', why does a games console need secondary storage? give an example, do not just describe what secondary storage is!

Answer

- To store data once the power is turned off / permanently // for non-volatile storage
- Examples could include:
 - Storing games
 - Save games
 - User data/profiles
 - Downloaded content/applications/updates

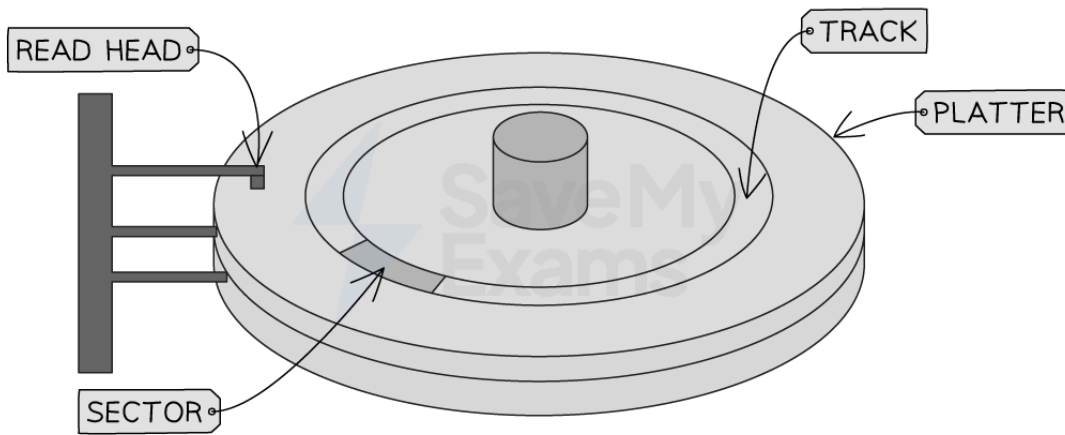
Magnetic Storage

What is magnetic storage?

- Magnetic storage is a type of **non-volatile media** that uses **magnets** (polarity) to store binary 0s and 1s
- A magnetic hard disk is made up of **several metal discs** coated with a magnetic material
 - These are called **platters**
 - Iron particles on each platter are **magnetised to represent a 0 or 1**

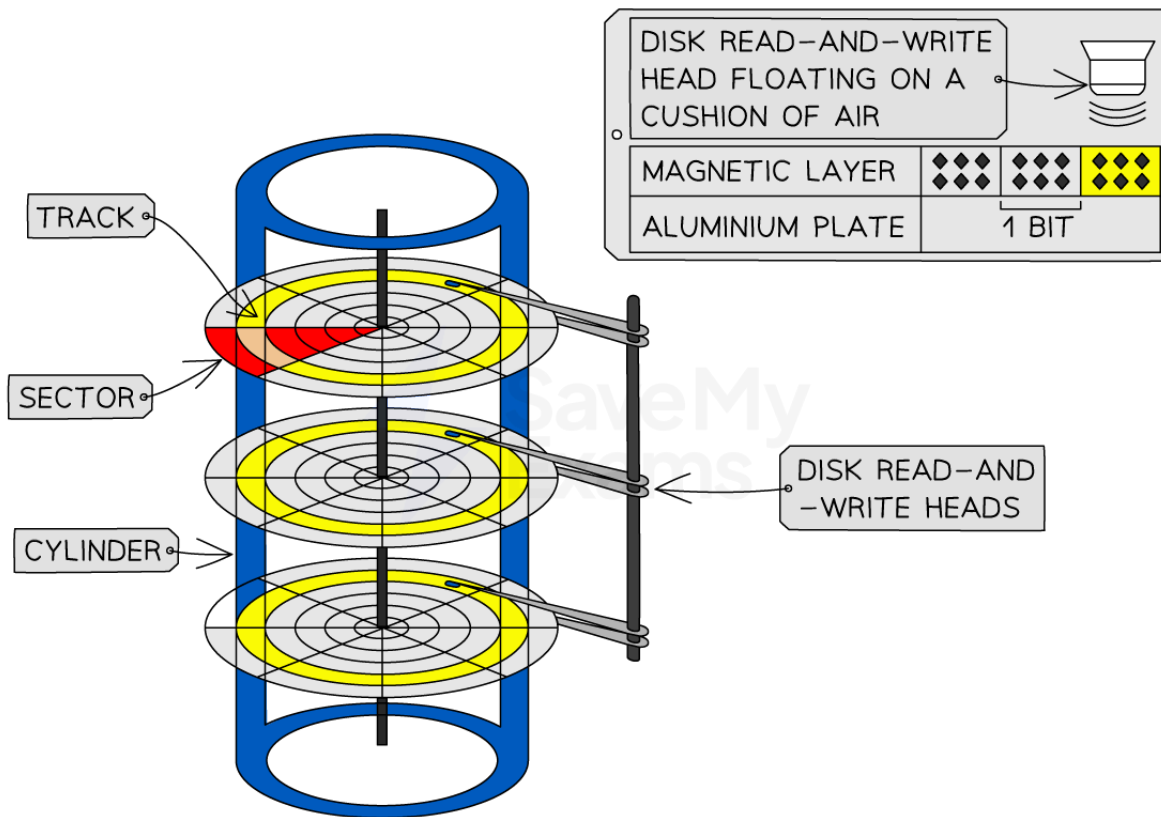


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- Each platter is divided by concentric circles creating several **tracks** and wedge shaped **sectors**
- Where they intersect is a **track sector**



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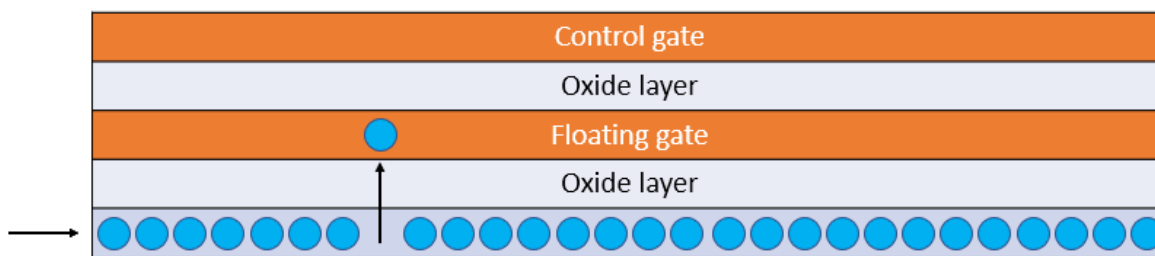
- The hard drive **spins the metal disk(s)** at a high speed (typically around 5400–7200 RPM) using a **motor**
- A **read/write arm**, controlled by an actuator, **moves the head** over the surface of the disc to the location of the data
- The data is read/written using **electromagnets**

Advantages	Disadvantages
Capacity - High storage	Durability - Moving parts can get damaged if dropped
Cost - Low per gigabyte	Portability - Heavy & bulky making them less convenient for transport
Speed - Moderate read/write access	Reliability - Prone to mechanical failure
	Noise - Loud (spinning disks)

Solid-state (Flash Memory)

What is solid state (flash memory)?

- Solid-state (flash memory) storage is a type of **non-volatile media** that uses **electronic circuits** to store binary 0s and 1s
- Examples of solid state storage devices include:
 - **Solid state drives (SSD)** - replacing magnetic hard drives as a computers primary secondary storage device as capacity increases and cost decreases
 - **USB flash memory** - most common form of portable storage device
- It uses **NAND** and **NOR gates** in electrical circuits to **persistently** control the flow of electrons





Your notes

- Solid-state (flash memory) is memory made up of **tiny cells** that can contain **one bit of data** (1 or 0)
- Each cell **contains a transistor** that acts as a **switch** that can be **turned on or off**
- The transistor contains **two** main parts:
 - **Control gate** - top layer of transistor, connects to circuit and controls if current can flow through the transistor
 - **Floating gate** - can hold a charge (like a tiny rechargeable battery) and is sandwiched between two layers of insulating material (Oxide)
- To **store data** a charge is placed on the floating gate
 - A **high voltage is applied** to the **control gate**, which allows electrons to be **pushed through the oxide layer** and onto the **floating gate**
- To **remove data**, a high voltage charge is applied in the **opposite direction**, **pulling the electrons off the floating gate**

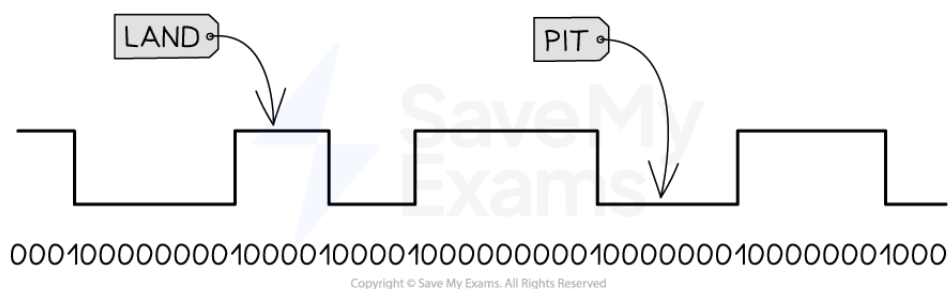
Advantages	Disadvantages
Capacity - Medium/high storage	Cost - Very high per gigabyte
Speed - Very fast read/write access	Reliability - Limited read/write cycles
Durability - No moving parts	
Portability - Small and no moving parts	
Noise - Silent	

Optical Storage

What is optical storage?

- Optical storage is a type of **non-volatile media** that uses **lasers** to burn the surface of a disk, creating pits and lands suitable for storing binary 0s and 1s
- Examples of optical storage include:
 - **Blu-rays** have the largest capacity
 - **CDs** have the lowest capacity

- **CD-R** are read-only (you cannot save data on to them)
- **CD-RW** can be written to and read from
- **DVD-RW** can be written to and read from



- All optical devices work by **shining a laser at the disk** and **processing the reflection**
- An arm moves the laser **across the surface of the disk**
- In **CD-Rs** a laser burns the data, **permanently** on to the disk, by creating pits and lands
- The laser is also used to **read the data from the pits and lands**
- When the laser light **hits the point where the pit changes into a land** or vice versa the **light scatters** and **is not reflected back as well**.
- This is **captured by a sensor** and can be **interpreted as a change in the binary value**

Advantages	Disadvantages
Cost - Very low per gigabyte	Capacity - Very low
Durability - No moving parts	Speed - Very slow read/write access
Portability - Small and no moving parts	Reliability - Prone to scratches
Noise - Silent	



Worked Example



Your notes

Zarmeen has purchased a new tablet computer. The tablet has an internal secondary storage device.

The storage device is a solid state device.

1. Give **three** benefits of the tablet having a solid state device instead of a magnetic device **[3]**
2. Give **two** drawbacks of the tablet having a solid state device instead of a magnetic device **[2]**

Answers

- Benefits
 - Faster access / read/write speeds
 - Smaller in physical size
 - More durable // no moving parts
- Drawbacks
 - More expensive (per gigabyte)
 - Smaller capacity (usually)
 - Limited number of read/write cycles

Guidance

- Portable is not enough on it's own
- No moving parts is not enough on it's own

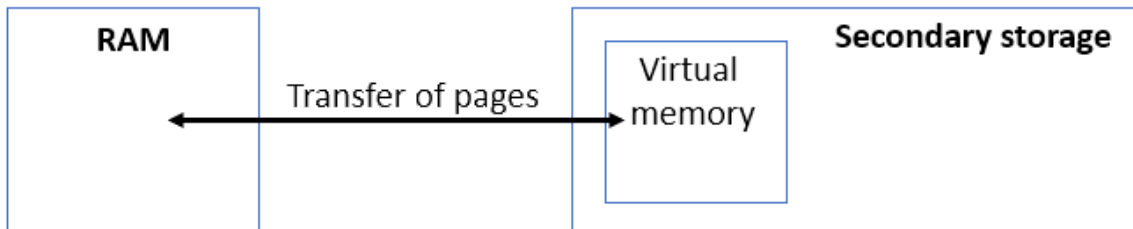


Your notes

Virtual Memory

Virtual Memory

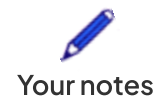
What is virtual memory?



- Virtual memory is an **extension of primary storage** (RAM) located on **secondary storage**
- In situations where RAM is close to being **full**, virtual memory can be used so that the computer **remains operational**
- Programs and data **not currently** being **executed** are transferred to virtual memory
- Programs and data are transferred back to RAM when they **are needed**
- Programs are stored as **pages of data**
- Pages** of data are **transferred** between RAM and virtual memory when needed
- Virtual memory is much **slower** than RAM and its use will impact negatively on system performance
- To avoid the use of virtual memory, **increase** the size of the **RAM**

Example

RAM (primary storage)		Virtual Memory (secondary storage)
Spreadsheet Software Operating System	↔	Video Editing Software



- In the example above, the computer is turned on and the elements of the operating system are loaded into RAM
- The user opens some spreadsheet and video editing software but RAM is full
- The program not currently being used (video editing) is transferred to virtual memory
- When the user wants to use the video editing software again, it is transferred back from virtual memory to RAM



Case Study

A student uses software to create 3D models.

- 3D modelling often means processing large amounts of data
- This takes up a large amount of RAM
- If there is not enough physical RAM, virtual memory is used to:
 - Extend the RAMs capacity
 - Ensure the software does not freeze/crash when RAM gets full



Worked Example

A computer only has 4GB of RAM, Fynn says that virtual memory can be used instead of adding more RAM.

1. Explain how virtual memory can compensate for the lack of RAM in Fynn's computer **[3]**
2. Explain why it would be beneficial for Fynn to get more RAM instead of relying on virtual memory **[2]**

How to answer these questions

1. What is the definition of virtual memory? how does it work?

2. What is the downside of using virtual memory? turn this into an advantage of adding more RAM

Answers

- How can Virtual Memory compensate
 - Virtual Memory is used when RAM is full
 - ...part of the secondary storage used as (temporary) RAM
 - Data from RAM is moved to the secondary storage/Virtual Memory (to make space in RAM)
 - RAM can then be filled with new data
 - When data in Virtual Memory is needed it is moved back to RAM
- Benefit of more RAM
 - More RAM will improve the performance of the computer // More RAM will speed up the access to data
 - Virtual Memory is slower to access than RAM direct (because it has to go back to RAM first)
 - Moving data between RAM and VM takes processor time



Your notes



Your notes

Cloud Storage

Cloud Storage

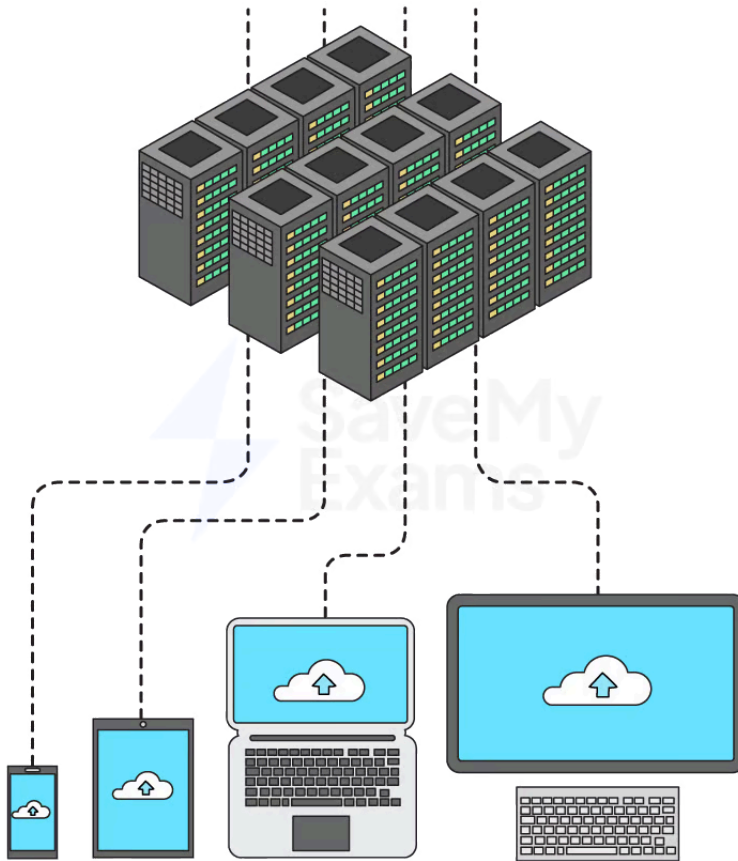
What is cloud storage?

- Cloud storage is a term to describe **long-term (secondary) storage of data** that resides in a **remote location**, accessible only **via a wide area network** (Internet)
- Data is stored on **remote servers**, typically using magnetic storage (**HDD**), but increasingly using solid state (**SSD**)

Advantages	Disadvantages
Reliability - Access data anywhere, any time, from any device	Cost - Typically a monthly/yearly subscription plan
Security - Automatic backups	Availability - No access without an internet connection
Capacity - Endless storage space (cost permitting), uses less space on own device	
Collaboration - Multiple people can access the same file at the same time	



Your notes



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Advantages of Cloud Storage

- Cloud storage offers several advantages over storing data on a local secondary storage device

Benefit	Explanation
Accessibility	- Data can be accessed from anywhere - Data can be accessed by anyone with the relevant permissions, making it quick to share files and collaborate with others - Data can be accessed on any device with an internet connection
Scalability	Cloud storage providers allow customers to increase or decrease their storage capacity as needed



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Reliability	Cloud storage providers often use multiple servers to store and backup data, reducing the risk of data loss due to hardware failure
Security	- Cloud storage providers offer advanced security features, such as data encryption and multi-factor authentication, to protect user data from unauthorised access - There is no need to hire specialist staff as IT services being provided by the cloud storage provider

- You also do not need to purchase local secondary storage devices
- One cloud storage centre is more **environmentally friendly** than millions of individual servers
- There are some drawbacks to using cloud storage
 - If the user has a slow or **unreliable internet connection** they would have problems **accessing** their files
 - If you are storing a lot of data it can be **expensive**
 - You have to pay the cloud storage provider
 - You may need to commit to an annual subscription
 - You may have to pay for high download and upload data transfer limits
 - Data being sent over the internet has the potential to be **intercepted**
 - Security is managed by the cloud storage provider which means the user **does not have complete control** over it but is still legally responsible for ensuring any **personal and sensitive data** is secure



Worked Example

A school currently stores all of their data on local servers. They are considering upgrading their system to utilise cloud storage.

Explain two disadvantages to the school of storing their data in the cloud.

[4]

- You need a constant internet connection [1]
- if the internet connection failed, teachers would not be able to access lesson files [1]
- if the internet connection failed, teachers would not be able to access student data/take electronic registers [1]



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- if the internet connection failed, students would not be able to access their work/coursework [1]
- Reliant on third party to carry out security procedures [1]
- but the school is still legally responsible if data is accessed by unauthorised people [1]
- Data stored on the cloud could be hacked or intercepted [1]

Explain two advantages to the school of storing their data in the cloud.

[4]

- Additional storage can be purchased if needed [1]
- The cloud hosting company provides back up [1]
- The cloud hosting company provides security [1]
- Data can be accessed from anywhere and from any device provided there is an internet connection [1]
- So teachers would be able to access data from home [1]
- So students would be able to access work from home [1]